Ellipse: Formula Sheet

Recognition and definition

Ellipse: $\Delta \neq 0,\ h^2 < ab$

Definition: $PS_1 + PS_2 = 2a$

Focus–directrix: $PS = e\,PM,\ 0 < e < 1$

Standard ellipse

Equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Eccentricity: $b^2 = a^2(1 - e^2)$

Foci, directrices: $(\pm ae, 0)$, $x = \pm\frac{a}{e}$

LR: $\frac{2b^2}{a}$, ends $\left(\pm ae, \pm\frac{b^2}{a}\right)$

Focal distances: $a \mp ex_1$

Vertical: foci $(0, \pm ae)$, directrices $y = \pm\frac{a}{e}$

Shifted:

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Alternate:

$$\frac{p_2^2}{a^2} + \frac{p_1^2}{b^2} = 1$$

Auxiliary circle, parametric form

Auxiliary circle: $x^2 + y^2 = a^2$

Point: $(a\cos\theta, b\sin\theta)$

Chord $\alpha\beta$:

$$\frac{x}{a}\cos\frac{\alpha+\beta}{2} + \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha-\beta}{2}$$

Focal chord:

$$\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{e-1}{e+1}$$

Position, tangency, director circle

Position: $S_1 < 0,\ = 0,\ > 0$

Tangency: $c^2 = a^2m^2 + b^2$

Line $lx + my + n = 0$: $a^2l^2 + b^2m^2 = n^2$

Contact point:

$$\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$$

Director circle: $x^2 + y^2 = a^2 + b^2$

Slopes from (h, k): $(h^2 - a^2)m^2 - 2hkm + k^2 - b^2 = 0$

Tangent and normal

Tangent at point:

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

Tangent at θ:

$$\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$$

Slope form: $y = mx \pm \sqrt{a^2m^2 + b^2}$

Normal at point:

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$$

Normal at θ:

$$ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$$

Normal condition:

$$c^2 = \frac{m^2(a^2-b^2)^2}{a^2 + b^2m^2}$$

Tangents at α, β meet:

$$\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{b\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$$

Chords

Mid-point chord: $T = S_1$

Chord of contact / polar: $T = 0$

Pair of tangents: $SS_1 = T^2$

Pole of $lx + my + n = 0$:

$$\left(-\frac{a^2l}{n}, -\frac{b^2m}{n}\right)$$

Diameter: $y = -\frac{b^2}{a^2m}x$

Highlights

Focal sum: $PS_1 + PS_2 = 2a$

Perpendiculars from foci: $p_1p_2 = b^2$

Their feet: on $x^2 + y^2 = a^2$

Tangent–directrix part: $90°$ at focus

Least tangent-triangle area: ab